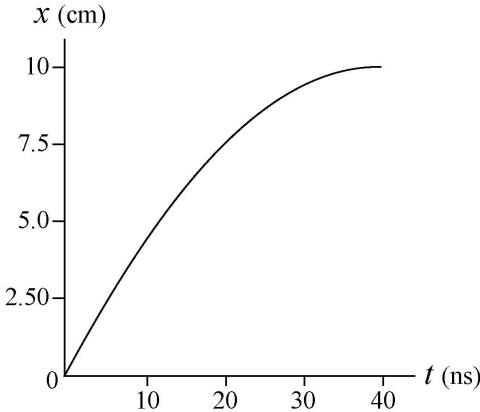
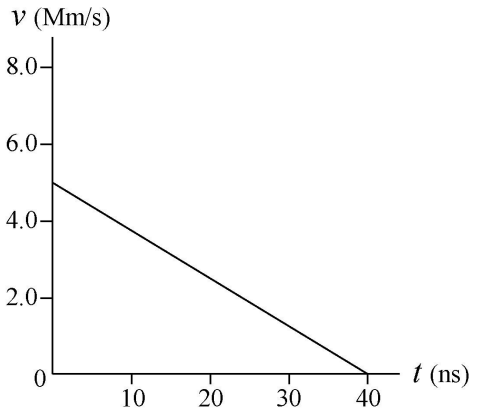
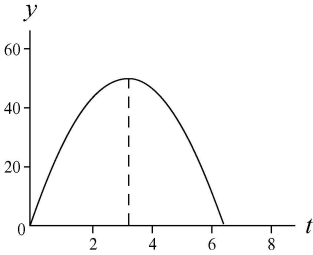
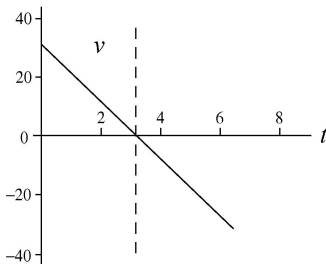
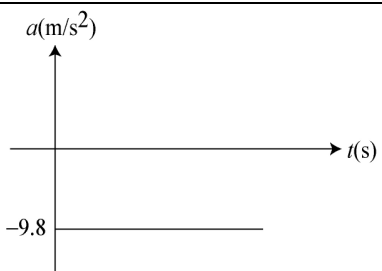
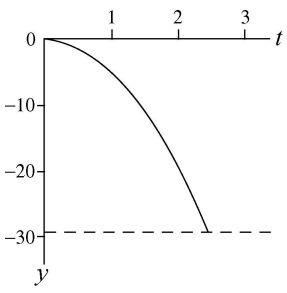
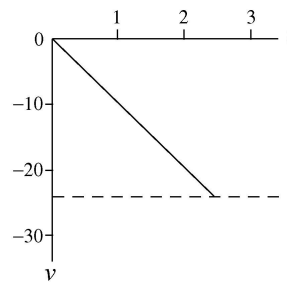
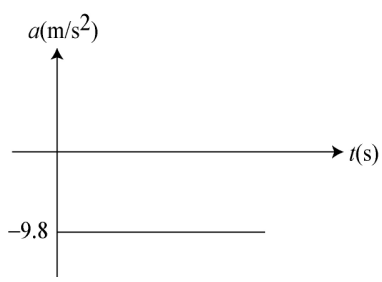


Problems for Ch 2 HRW – Motion in a straight line
 Addisionele Probleme vir Hfst 2 HRW – Beweging in 'n reguit lyn

	9 th Edition	8 th Edition
Questions		
Problems	1, 11, 15, 18, 19, 22, 23, 26, 27, 29, 37, 41, 44, 45, 47, 55, 74, 78, 99	3, 9, 15, 20, 19, 21, 27, 24, 23, 31, 35, 39, 48, 47, 45, 55, 80, 86, 108

1	3	13 m.	
11	9	128 km/h is needed.	
15	15	(a) At $t = 1$ s, the velocity is $v = (-12 + (6)(1)) = -6$ m/s. (b) Since $v < 0$, it is moving in the $-x$ direction at $t = 1$ s. (c) At $t = 1$ s, the <i>speed</i> is $ v = 6$ m/s. (d) For $0 < t < 2$ s, $ v $ decreases until it vanishes. For $2 < t < 3$ s, $ v $ increases from zero to the value it had in part (c). Then, $ v $ is larger than that value for $t > 3$ s. (e) Yes, since v smoothly changes from negative values (consider the $t = 1$ result) to positive. (f) No. In fact, from $v = -12 + 6t$, we know that $v > 0$ for $t > 2$ s.	
18	20	(a) At $t = 3.5$ s $x(3.5) = 61$ m . (b) $v(t) = 24t - 6t^2$ At $t = 3.5$ s yields $v(3.5) = 11$ m/s. (c) $a(t) = 24 - 12t$ At $t = 3.5$ s, $a(3.5) = -18$ m/s ² . (d) $x = 64$ m (e) From (d), we see that the x reaches its maximum at $t = 4.0$ s. (f) $v_{\max} = 24$ m/s. (g) $t = 24/12 = 2.0$ s. (h) $a = -24$ m/s ² . (i) $v_{\text{avg}} = \frac{54-0}{3-0} = 18$ m/s.	(a) At $t = 3.0$ s $x(3.0) = 54$ m . (b) $v(t) = 24t - 6t^2$ At $t = 3.5$ s yields $v(3.5) = 18$ m/s. (c) $a(t) = 24 - 12t$ At $t = 3.0$ s, $a(3.0) = -12$ m/s ² . (d) $x = 64$ m (e) From (d), we see that the x reaches its maximum at $t = 4.0$ s. (f) $v_{\max} = 24$ m/s. (g) $t = 24/12 = 2.0$ s. (h) $a = -24$ m/s ² . (i) $v_{\text{avg}} = \frac{54-0}{3-0} = 18$ m/s.
19	19	The average acceleration has magnitude 20 m/s^2 and is in the opposite direction to the particle's initial velocity.	
22	21	(a) unit of c must be that of length/time ² , or m/s ² in the SI system. (b) b must have a unit of length/time ³ , or m/s ³ . (c) $t = 1.3$ s . (d) The total path length it travels is $1.3 \text{ m} + 1.3 \text{ m} + 64 \text{ m} = 67 \text{ m}$. (e) $\Delta x = -64$ m . (f) $v(1.0) = 2.0$ m/s (g) $v(2.0) = -8.0$ m/s (h) $v(3.0) = -30$ m/s (i) $v(4.0) = -64$ m/s (j) $a(1.0) = -4.0$ m/s ²	(a) unit of c must be that of length/time ² , or m/s ² in the SI system. (b) b must have a unit of length/time ³ , or m/s ³ . (c) $t = 1.0$ s . (d) The total path length it travels is $1.0 \text{ m} + 1.0 \text{ m} + 80 \text{ m} = 82 \text{ m}$. (e) $\Delta x = -80$ m . (f) $v(1.0) = 0.0$ m/s (g) $v(2.0) = -12.0$ m/s (h) $v(3.0) = -360$ m/s (i) $v(4.0) = -72$ m/s (j) $a(1.0) = -6.0$ m/s ²

		(k) $a(2.0) = -16 \text{ m/s}^2$ (l) $a(3.0) = -28 \text{ m/s}^2$ (m) $a(4.0) = -40 \text{ m/s}^2$	(k) $a(2.0) = -18 \text{ m/s}^2$ (l) $a(3.0) = -30 \text{ m/s}^2$ (m) $a(4.0) = -42 \text{ m/s}^2$
23	27	$a = 1.62 \times 10^{15} \text{ m/s}^2$.	
26	24	(a) $x = 0.144 \text{ m}$. (b)  $x \text{ (cm)}$ $t \text{ (ns)}$  $v \text{ (Mm/s)}$ $t \text{ (ns)}$	
27	23	(a) $v = 16 \text{ m/s}$ (b) $v = 18 \text{ m/s}$	
29	31	(a) $\Delta x = 10.59 \text{ m} \approx 10.6 \text{ m}$. (b) $t_1 = 4.17 \text{ s} = t_2$ $t_3 = 33.21 \text{ s}$ Therefore, the total time is $8.33 \text{ s} + 33.21 \text{ s} \approx 41.5 \text{ s}$.	
37	35	(a) $a = 4.0 \text{ m/s}^2$. (b) acceleration vector points in the $+x$ direction	
41	39	40 m.	
44	48	(a) $v_o = 3.77 \text{ m/s}$ (b) $v = 1.81 \text{ m/s}$ (c) The armadillo goes 0.167 m higher.	(a) $v_o = 3.70 \text{ m/s}$ (b) $v = 1.74 \text{ m/s}$ (c) The armadillo goes 0.1547 m higher.
45	47	(a) $v_o = 31.3 \text{ m/s}$ (b) $T = 6.4 \text{ s}$ (c) SI units are understood in the x and v graphs shown. The acceleration graph is a horizontal line at -9.8 m/s^2 .  y t  v t	

			
47	45	(a) $\Delta y = 29.4 \text{ m}$ (b) $t = 2.45 \text{ s}$ (c) SI units are used in the graphs, and the initial position is taken as the coordinate origin. The acceleration graph is a horizontal line at -9.8 m/s^2.   	
55	55	(a) $a_{avg} = 857 \text{ m/s}^2$ (b) upward	
74	80	$t = 0.000310 \text{ h} \approx 1.1 \text{ s}.$	$t = 0.000358 \text{ h} \approx 1.3 \text{ s}.$
78	86	We take the moment of applying brakes to be $t = 0$. The deceleration is constant so that Table 2-1 can be used. Our primed variables (such as $v'_0 = 82 \text{ km/h} = 23 \text{ m/s}$) refer to one train (moving in the $+x$ direction and located at the origin when $t = 0$) and unprimed variables refer to the other (moving in the $-x$ direction and located at $x_0 = +950 \text{ m}$ when $t = 0$). We note that the acceleration vector of the unprimed train points in the <i>positive</i> direction, even though the train is slowing down; its initial velocity is $v_0 = -144 \text{ km/h} = -40 \text{ m/s}$. Since the primed train has the lower initial speed, it should stop sooner than the other train would (were it not for the collision). Using Eq 2-16, it should stop (meaning $v' = 0$) at $x' = \frac{(v')^2 - (v'_0)^2}{2a'} = \frac{0 - (23 \text{ m/s})^2}{-2 \text{ m/s}^2} = 265 \text{ m}.$ The speed of the other train, when it reaches that location, is	
		$x' = 200 \text{ m}.$ $v = 10 \text{ m/s}$ A concern that can be brought up is whether the primed train collides before it comes to rest; this can be studied by computing the time it stops (Eq. 2-11 yields $t = 20 \text{ s}$) and seeing where the unprimed train is at that moment (Eq. 2-18 yields $x = 350 \text{ m}$, still a good distance away from contact).	

		$v = \sqrt{v_0^2 + 2a\Delta x} = \sqrt{(-40 \text{ m/s})^2 + 2(1.0 \text{ m/s}^2)(265 \text{ m} - 950 \text{ m})}$ $= 15 \text{ m/s}$ using Eq 2-16 again. Specifically, its velocity at that moment would be -15 m/s since it is still travelling in the $-x$ direction when it crashes. If the computation of v had failed (meaning that a negative number would have been inside the square root) then we would have looked at the possibility that there was no collision and examined how far apart they finally were. A concern that can be brought up is whether the primed train collides before it comes to rest; this can be studied by computing the time it stops (Eq. 2-11 yields $t = 23 \text{ s}$) and seeing where the unprimed train is at that moment (Eq. 2-18 yields $x = 295 \text{ m}$, still a good distance away from contact).	
99	108	$ v = 22.0 \text{ m/s}.$	